Stablecoin StressBench: Cross-Mechanism Transfer Learning for Executable Arbitrage Under a $12\times$ Optical Gap

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Abstract

Price charts show twelve times more profitable stablecoin arbitrage than a trader can capture, so a model that scores well on price labels can still lose money on every trade. We study when a learned trading model transfers across financial crises, using stablecoin depegs as the testbed. The central obstacle is distributional: a model trained on calm or single-regime data cannot identify the executable minority in a stress regime it has never seen, and every calm-trained family we test—tabular classifiers, sequence models, reinforcement learning, and changepoint detectors—fails economically despite high AUROC. Our main result is that *cross-mechanism transfer* solves this: a meta-labeling classifier trained on an algorithmic collapse (Terra/LUNA) and applied unchanged to a bank-reserve shock (USDC/SVB) recovers +82.5 bps (51.0% of a hindsight oracle), because depth withdrawal and spread widening are mechanism-invariant—the order book responds to uncertainty, not its cause. We further isolate *supervision format* as the binding constraint: at identical positive-label density, explicit binary labels earn +82.5 bps while reward optimisation (PPO-GRU) earns −29.2 bps. To support these findings we release Stablecoin StressBench: an 18-event catalogue across seven mechanism classes, per-minute VWAP (volume-weighted average price) book-walking labels with route provenance, and a hindsight oracle ceiling, under CC-BY 4.0.

Keywords

transfer learning, meta-labeling, reinforcement learning, execution-aware labels, market microstructure, stablecoin arbitrage, financial benchmarks

ACM Reference Format:

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1 Introduction

Stablecoin stress is not one phenomenon but seven. From 2020 to 2023 we catalogue 18 distinct depeg episodes spanning algorithmic death spirals (Terra/LUNA, IRON/TITAN), fiat-reserve shocks (USDC/SVB), regulatory winddowns (BUSD), exchange-credit collapses (FTX, Celsius), DeFi pool imbalances (Curve/USDT), collateral liquidations (DAI Black Thursday), and real-world-asset failures (USDR, Acala). Yet every prior study examines a single mechanism, and none asks the question a trader actually faces: of the price dislocations that appear during stress, how many can be captured at a profit?

The answer is far fewer than the chart suggests. When USDC slipped below \$1 during the March 2023 SVB crisis, Bitcoin bought with USDC looked cheaper than the same Bitcoin priced in real dollars—free money on a price chart. In practice it rarely was. By the time an order walked the book, the discount had narrowed; taker fees and settlement delay consumed the rest. Only 2.88% of one-minute windows survived all three costs, against 34.3% that appeared dislocated: a 12× gap between visible and executable arbitrage.

We separate three layers of the same event. **Layer 1 (optical)** is what the price chart shows. **Layer 2 (executable)** is what survives VWAP book-walking, fees, and settlement. **Layer 3 (captured)** is what a model can flag in advance. Gap 1 is the distance from Layer 1 to Layer 2; Gap 2, from Layer 2 to Layer 3. We call Layer-1-only dislocations ***optical***—real on the chart, illusory in the order book—and the survivors ***executable***. The two gaps have distinct causes. Gap 1 is a measurement problem: price labels overstate tradeable performance by a factor of twelve. Gap 2 is a training problem: no model trained on calm data can find the executable minority in a stress regime it has never seen. Figure 2 shows all three layers for the SVB test window and traces one representative false-positive minute from apparent profit to net loss. Basis points (bps) are hundredths of a percent (100 bps = 1%).

We establish each gap against a ground truth, mirroring the sanity-check-then-hard-case logic of controlled studies. Terra/LUNA, a mechanistically independent algorithmic collapse, serves as our known reference: it reproduces the optical-to-executable gap at 5.9× (2.40% executable) and supplies a clean cross-mechanism training signal. USDC/SVB, a fiat-reserve shock with route-complete order-book depth, is the execution-grade test case, where the gap reaches 12× (2.88% executable) and widens to 14× under a 20% depth haircut. That the same gap appears in two unrelated mechanisms tells us the phenomenon is structural, not an artefact of either event; only the Tier-A execution labels are SVB-specific.

The one approach that crosses zero exploits a simple regularity: the order book responds to uncertainty, not to its cause. Depth withdrawal and spread widening look identical in an algorithmic collapse and a bank-reserve shock, so a meta-labeler trained on Terra/LUNA recognises the executable signature in SVB without ever seeing it. This single transfer recovers half the return available to a hindsight oracle.

Contributions.

(1) **Cross-mechanism transfer for executable arbitrage.** A meta-labeling classifier trained on one crisis type and applied unchanged to another recovers +82.5 bps on SVB (51.0% oracle capture); four-event pooling reaches 51.8%. Transfer succeeds because the microstructure stress signature—depth withdrawal, spread widening—is mechanism-invariant, which SHAP attribution and matched depth-withdrawal curves confirm. (§6.3)

(2) **Supervision format, not label density, as the binding constraint.** At identical 17% positive-label density, explicit binary

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supervision earns +82.5 bps while reward optimisation (a conditioned PPO-GRU) earns −29.2 bps, locating the difference in how the learning signal is delivered rather than in data scarcity. Every other calm-trained family—tabular, sequence, class-balanced, regression, and changepoint—also fails, ruling out architecture and features as the remedy. (§6.2, 6.3.2)

(3) **An execution-aware benchmark that makes these claims measurable.** StressBench supplies the apparatus: an 18-event source-tracked catalogue across seven mechanism classes, per-minute VWAP book-walking labels with route provenance, and a hindsight oracle ceiling. It exposes a 12× optical-to-executable gap, so a model scored on price labels overstates tradeable performance by roughly twelve times—an evaluation-integrity result for any AI trading model on this data. (§3–5, 6.1)

2 Related Work

Stablecoin stress and depeg detection. Uhlig [1] and Griffin and Shams [2] analyse Terra/LUNA and stablecoin non-fundamental demand. Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse without modelling CEX execution costs. Cintra and Holloway [9] detect the Terra depeg 12 hours early via BOCPD on AMM reserves. We replicate their approach on CEX cross-quote data and find AUROC 0.229, confirming that AMM and CEX signals differ fundamentally. Wu and Liu [20] study tail spillovers across stablecoin designs without per-minute execution analysis. None of these works provide VWAP-grade execution labels, a hindsight oracle, or cross-mechanism transfer evaluation.

Limits to arbitrage and execution frictions. Shleifer and Vishny [4], Gromb and Vayanos [5], and Makarov and Schoar [7] show that visible price gaps need not be profitable once execution frictions are included. Our 12× ratio is this effect measured at one-minute resolution. Hautsch et al. [3] quantify settlement latency as a first-order cost on blockchain assets. Cont and Kukanov [6] establish VWAP book-walking as the correct profitability measure. Gogol et al. [18] measure Maximal Arbitrage Value on AMM-to-CEX routes. Their partial execution labels cover AMM venues only and lack a hindsight oracle. Adams et al. [19] study swap costs on Uniswap without stress-event execution analysis.

ML and RL for execution. Lopez de Prado [8] introduces meta-labeling, which we extend to the cross-mechanism transfer setting. Moallemi and Wang [16] and Ning et al. [17] supply the finite-horizon MDP framing we adopt. Mohl et al. [22] and Olby et al. [23] operate in reactive LOB environments with market-impact feedback. Our diagnostic uses historical replay without feedback, isolating training supervision as the variable.

3 Benchmark Design

3.1 Data Sources and Instruments

The 18-event catalogue defines the historical stress universe. The six-window empirical panel (56,134 one-minute bars) defines the model dataset, and the Tier-A SVB windows define the execution-grade testbed. Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs across two centralised exchanges (CEXs, i.e., Binance and Coinbase): Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference). The BTC-USDC buy leg uses kline-proxy depth for the SVB window because the perpetual contract did not exist on Binance until January 2024. A 20% depth haircut confirms robustness (gap widens to 14×, oracle stays above +130 bps).

Data quality varies by leg, and every record carries a provenance tag (Table 1).

Table 1: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	partial/proxy; real L2 where archived	real L2 (futures)
Sell BTC-USDT	raw/futures depth	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

Table 2: Dataset splits (56,134 one-minute bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

3.2 Event-Based Split Design

Splits are defined by event identity, so no model trained on the calm split has encountered stress dynamics. Threshold calibration uses the Terra/LUNA validation split, an independent stress event, so the SVB test split is never observed during model development. The 2.88% positive rate in the test split implies severe class imbalance. We therefore report economic net bps as the primary criterion. AUROC and AUPRC are secondary metrics.

Five leakage checks confirm validity: label shuffles produce zero or negative P&L; one-minute lookahead does not improve performance; the calm training split contains zero executable windows. Train and test are separated by event identity, and cross-mechanism calibration uses only the Terra/LUNA calibration block.

3.3 Historical Catalogue: Situating SVB in the Stress Universe

The execution-grade analysis centers on USDC/SVB. The 18-event catalogue motivates this choice and justifies cross-mechanism training: depth withdrawal and spread widening recur across all mechanism classes, so a model that has seen them in one event can recognise them in another.

Table 3 lists all 18 events with source-verified statistics ("est." denotes values from verified secondary sources, not per-minute data). Tier-A events support VWAP execution labels and oracle bounds. Tier-B events support price/liquidity claims. Tier-C events contribute taxonomy coverage. The pooled transfer experiment draws from Terra/LUNA (algorithmic), Celsius/3AC and FTX (exchange credit), and BUSD (regulatory), all held mechanistically distinct from the Tier-A SVB test event.

Figure 1 shows the event universe as a bubble timeline: mechanism class on the vertical axis, date on the horizontal, and bubble area proportional to the absolute maximum depeg.

Table 3: StressBench event catalogue: all 18 events, chronological. Max depeg in bps (+=above peg; †=percentage-point collapse to zero). "est." = source-verified estimate, not per-minute data. Tier A = execution-grade (VWAP labels + oracle); Tier B = price/liquidity claims; Tier C = taxonomy only.

Event	Date	Coin	Mechanism	Max dep.	Dur.	T	Event	Date	Coin	Mechanism	Max dep.	Dur.	T
DAI Black Thursday	Mar '20	DAI	Collateral	est. +150	days	B	Acala aUSD Exploit	Aug '22	aUSD	RWA/Niche	est. −99pp†	days	C
FEI Launch Stress	Apr '21	FEI	Algo./Reflex.	est. −20pp†	days	C	HUSD Issuer Failure	Aug '22	HUSD	Exch. Credit	est. −800	days	B
IRON/TITAN Collapse	Jun '21	IRON	Algo./Reflex.	est. −100pp†	hrs	C	Binance USDC⟶BUSD Conv.	Sep '22	BUSD	Regulatory	est. +0	wks	C
MIM/Wonderland Shock	Jan '22	MIM	Algo./Reflex.	est. −300	days	C	FTX Collapse	Nov '22	USDT	Exch. Credit	est. −50	days	B
Terra/UST Collapse	May '22	UST	Algo./Reflex.	est. −100pp†	wks	B	BUSD Regulatory Winddown	Feb '23	BUSD	Regulatory	est. −30	wks	B
Curve 3Pool/UST Stress	May '22	UST	DeFi Pool	est. −500	days	B	USDC/DAI DeFi Secondary	Mar '23	USDC	DeFi Pool	est. −200	days	B
Celsius/3AC Contagion	Jun '22	USDT	Exch. Credit	est. −100	wks	B	**USDC/SVB Stress**	Mar '23	USDC	Fiat Reserve	−13pp†	days	**A**
USDD/TRON Stress	Jun '22	USDD	Algo./Reflex.	est. −200	days	B	USDC Recovery (Post-SVB)	Mar '23	USDC	Fiat Reserve	−10	wks	**A**
							USDT/Curve Pool Stress	Jun '23	USDT	DeFi Pool	est. −80	days	B
							USDR RWA Failure	Oct '23	USDR	RWA/Niche	est. −50pp†	days	B

2A, 11B, 5C events. Tier-A VWAP labels available only for USDC/SVB (primary benchmark) and its recovery window.

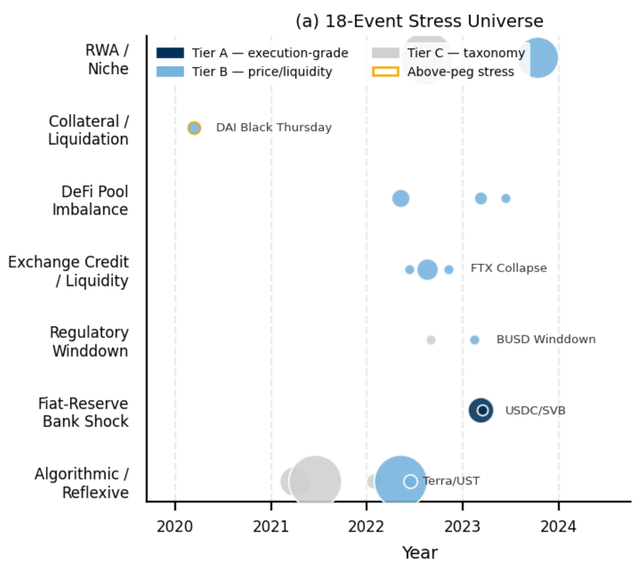

Figure 1: 18-event stress universe. Each bubble is one event; area ∝ |max depeg|; gold edge indicates above-peg stress. Tier A (navy); Tier B (blue); Tier C (grey). Basis fingerprints for Terra/LUNA and USDC/SVB appear in Figure 7.

Table 4: Comparison with related benchmarks. ✓ full; ∼ partial; — absent.

Work	Stress	Exec. labels	L2 prov.	Oracle	Cross-event
Cintra & Holloway [9]	✓	—	AMM	—	✓
Hernandez Cruz et al. [21]	✓	—	DEX	—	—
Gogol et al. [18]	—	∼	CEX	MAV	—
Adams et al. [19]	—	∼	CEX	—	—
StressBench (ours)	✓	✓	CEX	✓	✓

3.4 Benchmark Positioning

Table 4 positions StressBench against closest prior work. No prior work provides per-minute executable CEX labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation together.

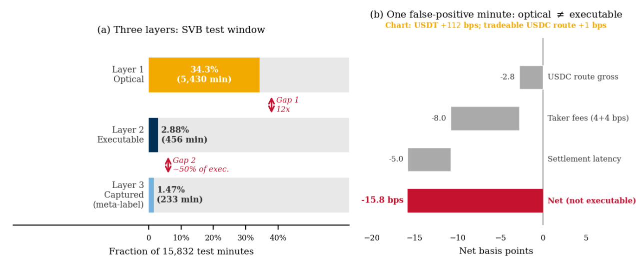

Figure 2: (a) Three-layer framework for the SVB test window (15,832 min). Layer 1 optical (34.3%), Layer 2 executable (2.88%), Layer 3 captured by cross-mechanism meta-labeling (1.47%). Gap 1 (12×) is a measurement problem; Gap 2 is a training problem. (b) Worked example: a minute that appears profitable on the chart nets −15.8 bps after VWAP book-walking, fees, and route-direction mismatch.

3.5 Feature Sets

Four nested sets isolate marginal signal value. `price_only` provides 5 cross-quote basis columns. `price_plus_book` adds 7 microstructure signals (spread, depth, imbalance). `price_book_frag` adds 3 cross-venue fragmentation metrics. `price_book_settle` adds 7 on-chain settlement proxies, which are kline-proxy approximations for the SVB and Terra/LUNA windows (Tier B).

4 Execution-Aware Labels

4.1 Three-Layer Framework

Figure 2 structures the paper's central argument visually. Layer 1 (optical) contains every minute where the price chart shows a profitable gap. Layer 2 (executable) contains the minutes that remain profitable after VWAP book-walking, taker fees, and settlement latency. Layer 3 (captured) contains the minutes a deployed model can identify in advance. Gap 1 is the distance from Layer 1 to Layer 2. Gap 2 is the distance from Layer 2 to Layer 3. Both gaps are nonzero, independent, and large. Panel (b) of Figure 2 traces one representative false-positive minute through the full cost stack, showing why 34.3% optical activity collapses to 2.88% executable.

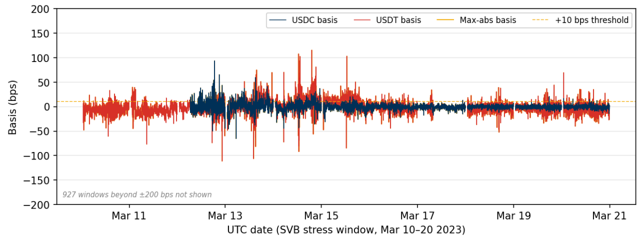

Figure 3: Cross-quote basis during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes above 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution ratio is $12\times$.

4.2 Cross-Quote Basis and Labels

The cross-quote basis is the percentage gap between BTC priced in a stablecoin (e.g., BTC-USDC) and BTC priced in real dollars (BTC-USD on Coinbase). A non-zero basis signals an apparent dislocation. Three equations define the label construction.

$$b_{\mathrm{USDC}}(t) \;=\; 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad [\mathrm{bps}] \tag{1}$$

$$b_{\max}(t) \;=\; \max(|b_{\mathrm{USDC}}(t)|,\; |b_{\mathrm{USDT}}(t)|) \tag{2}$$

$$y_{\mathrm{basis}}(t) \;=\; \mathbf{1}[|b_{\mathrm{USDC}}(t{+}h)| > \tau] \tag{3}$$

Primary task: `basis_usdc_1m_gt10bps` (τ=10 bps, h=1 min).

4.3 VWAP Execution and Net Profit Labels

$$\mathrm{net}(q,t) \;=\; 10{,}000 \times \left[\frac{\mathrm{VWAP}_{\mathrm{sell}} - \mathrm{VWAP}_{\mathrm{buy}}}{P_{\mathrm{ref}}} - f_{\mathrm{buy}} - f_{\mathrm{sell}} - \delta_{\mathrm{settle}} \right] \quad [\mathrm{bps}] \tag{4}$$

$$y_{\mathrm{arb}}(q,h,t) \;=\; \mathbf{1}\left[\max_{s \in [t,\, t+h]} \mathrm{net}(q,s) \;>\; c \right] \tag{5}$$

Profit threshold c=10 bps, taker fee 4 bps per side, and δ=5 bps. The δ assumption is validated against 100K on-chain USDC ERC-20 transfers: median cost was 1.4–3.2 bps on nine of ten SVB days. The one exception (Mar 11 gas spike) saw P95 reach 8.6 bps.

$$\mathrm{Oracle}(q) \;=\; \frac{1}{|\mathcal{T}^+|} \sum_{t \in \mathcal{T}^+} \mathrm{net}(q,t), \qquad \mathcal{T}^+ = \{t : \mathrm{net}(q,t) > c\} \tag{6}$$

The oracle is a hypothetical strategy that trades every profitable window with perfect hindsight and represents an upper bound no deployable model can exceed. Oracle capture $\kappa_M = \bar{r}_M/\mathrm{Oracle}(q)$ is the fraction of the oracle's per-trade return recovered by model M.

5 Models and Evaluation

5.1 Model Ladder

The ladder is a sequential elimination experiment with one result at every rung: every family fails for the same reason, distributional mismatch between calm training and stress test.

Architecture. No help. The GRU posts the highest AUROC among sequence models (0.80) and the worst economic return (−239 bps, Table 8). Discriminability does not translate to profit at 2.88% positive rate.

Feature engineering. No help. All four nested feature sets produce negative returns. Depth and settlement signals offer no lift when the training split contains zero stress-regime examples.

Class balancing. No help. Cost-sensitive reweighting cannot synthesise stress dynamics from calm data; the issue is distributional, not proportional.

Regression targeting. No help. ExpNetProfitRegressor optimises the economic criterion directly and reaches −73.8 bps. Label mismatch is not the explanation.

Regime detection. No help. BOCPD, CUSUM, and EWMA all fail because CEX basis is mean-reverting, not change-point structured (BOCPD AUROC 0.229).

RL reward optimisation. No help at identical density. A conditioned PPO-GRU at 17% primary-signal density earns −29.2 bps. The reward signal is not a substitute for explicit labels even when positive-example density is matched.

The cross-mechanism extension of meta-labeling operates in three steps. Meta-labeling [8] is a two-step filter: a cheap rule flags candidate minutes, then a learned classifier keeps only the ones likely to survive execution.

(1) **Primary filter.** Flag t if $|b_{\mathrm{USDC}}(t)| > \tau_1$ (τ_1=10 bps).
(2) **Secondary classifier.** Fit f_θ on $\{(x_t,\, y_{\mathrm{arb}}(t))\}_{t \in \mathcal{E}_{\mathrm{train}}}$, where $\mathcal{E}_{\mathrm{train}}$ is a mechanistically distinct stress event.
(3) **Entry.** Signal at t iff primary fires and $f_\theta(x_t) > \theta^*$, where $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i > \theta} \mathrm{net}(q, t_i)$ on the Terra/LUNA calibration block.

5.2 RL Execution Policy Baseline

The entry-timing problem is framed as a finite-horizon MDP. At each minute t the agent observes the 30-step lookback of `price_plus_book` features, selects $a_t \in \{\texttt{enter}, \texttt{wait}\}$, and receives reward $r_t = \mathrm{net}(q,t)$ on entry, 0 on wait. We train a PPO agent with a GRU policy network (64 units, 30-step lookback) using the cross-mechanism protocol: training on Terra/LUNA stress windows only and evaluating on USDC/SVB without retraining. This directly tests whether the sequential timing framing overcomes the distributional mismatch that defeats all calm-trained models.

5.3 Calibration and Evaluation

Net bps denotes mean net basis points per triggered trade after VWAP, taker fees, and settlement penalties. Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i > \theta} \mathrm{net}(q, t_i)$ subject to a minimum of 25 trades. Primary metrics are mean net bps per triggered trade and oracle capture κ_M. Secondary metrics are AUROC and AUPRC. NoTrade at 0 bps is the economic floor.

Positive density clarification. Three densities appear in the paper. 2.88% is the fraction of SVB test minutes that are executable. 2.30% is the fraction of the full Terra/LUNA stream that is executable. 17% is the fraction of primary-signal minutes (those where $|b_{\mathrm{USDC}}| > 10$ bps) that are executable in Terra/LUNA; this is the density seen by the conditioned PPO-GRU and the meta-labeler secondary classifier during training. The RL-vs-meta-labeling comparison holds density fixed at the 17% figure.

Table 5: Master oracle and gap table. Every oracle value cited in the paper appears here. Gap = oracle minus best deployable model.

Task / Window	N	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m` (full test)	15,832	+161.7 bps	0 bps[a]	162 bps	0.253
`exec_arb_q10k_5m` (full test)	15,832	+224.6 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m` (full test)	15,832	+146.8 bps	−112.7 bps	260 bps	0.063
SVB stress phase only (Mar 10–14)	7,189	+259.9 bps	–	–	–
Terra/LUNA validation split	11,526	+87.6 bps	–	–	–

[a]Best calibrated model is NoTrade (0 trades).
Kline-proxy buy leg; 20% haircut bounds in §3.1.

Table 6: Price-to-execution gap ($10K, 1-min horizon, SVB test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

Five evaluation tasks ship with StressBench v1.0 (train: calm or Terra block; val: Terra calibration; test: SVB): `optical_basis` (AUROC/AUPRC), `exec_arb_q10k` and `exec_arb_q50k` (net bps, oracle capture), `cross_mech` (net bps), and `policy_entry` (net bps, trades). Valid submissions must report net bps per trade, oracle capture, cost assumptions, notional size, and data tier used. Models are ranked by net bps, not AUROC.

6 Results

Table 5 maps every oracle figure in the paper to its task and window. All subsequent references point back to one of these rows.

6.1 The 12× Execution Barrier

At a 10 bps threshold, 34.3% of test minutes are optically dislocated but only 2.88% are executable after VWAP slippage, taker fees, and settlement latency. The ratio persists across all basis levels and widens with notional size: the executable rate falls from 2.88% at $10K to 0.03% at $500K (Figure 4). Book depth, not fees, prices out institutional participants.

Block-bootstrap CIs (60-min blocks, 2,000 resamples): executable rate 95% CI $[1.47, 4.59]$%; ratio $[7.6, 23.0]\times$; oracle $[+213.9, +304.2]$ bps.

Table 7 shows the stress/recovery split. All executable windows fall in the SVB stress phase. The recovery phase contains zero executable windows despite 21.6% optical activity, confirming that depth returns before the basis closes. Terra/LUNA independently replicates the gap at 5.9× (2.40% executable of 14.0% optical), confirming that the optical-to-executable phenomenon is not specific to reserve-bank shocks. The Tier-A execution-grade labels are one event; the underlying finding is two.

If only one minute in twelve is tradeable even with perfect hindsight, the next question is whether any model can find those minutes in advance.

The gap is robust across the full cost-parameter space: sweeping taker fees from 2 to 10 bps and settlement penalty from 0 to 10 bps moves the executable rate between 2.84% and 3.66%.

Table 7: Optical and executable diagnostics by window. SVB stress/recovery are Tier-A execution-grade. Terra/LUNA is validation-grade only.

Window	Min.	Optical	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	**0.00%**	—
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

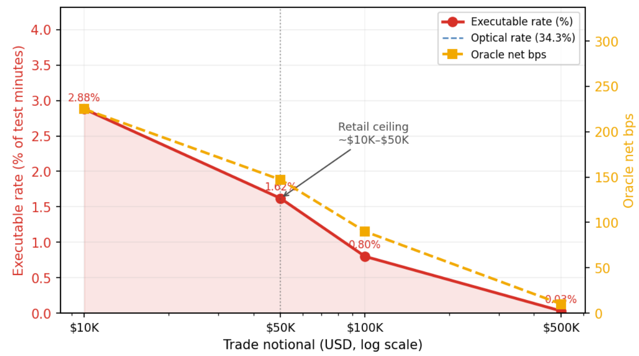

Figure 4: Notional scaling (SVB test split). The executable rate collapses from 2.88% at $10K to 0.03% at $500K: book depth, not fees, is the binding institutional constraint.

6.2 Calm-Training Failure

The central finding is not that individual models fail, but that all six families converge to the same result: no architecture overcomes the distributional mismatch between calm training and stress test. This convergence across every model family tested is itself the diagnostic: the failure is structural, not a gap in model coverage. Table 8 shows the tabular model results on `basis_usdc_1m_gt10bps`; oracle gap values map to Table 5.

Price-threshold rules overtrade massively (607–1,969 trades at −136 to −347 bps). False positives are not caused by thin books. True positives and false positives carry virtually identical book depth (72,805 vs. 59,961 BTC units on average), so depth is not the differentiator. False positives are caused by route-direction mismatch: roughly a third of them fire on USDT dislocations while the USDC route is simultaneously non-executable. True-positive windows carry a mean USDC basis of 344 bps. False positives carry just 0.56 bps, indicating the primary signal fired on the wrong stablecoin pair.

The discrimination–profitability disconnect. Among sequence models, the GRU achieves AUROC 0.80 (the highest) yet earns −239 bps (the worst return). AUROC (area under the ROC curve) measures how well a model ranks all windows by probability of profit, but even perfect ranking is worthless if no threshold can isolate the 2.88% executable minority. Table 8 shows this pattern holds across every family: high AUROC does not imply positive returns. The MLP-Window reaches −341 bps. A backtest reporting AUROC or F1 on basis labels overstates tradeable performance by roughly 12×. Any capital allocation based on such scores should be discounted accordingly.

Figure 6 decomposes the 260 bps oracle gap (Table 5) into three measurable components. False-positive exposure is the dominant

Table 8: Tabular model ladder (`basis_usdc_1m_gt10bps`). Hit% is fraction of executable trades. †Oracle ceiling. *0-trade degenerate.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	–	+161.7	316	–	100%
`lgbm`*	all	–	0	0.653	–
`logistic`	all	−49.1	1	0.133	0%
`gross_arb`	px	−136.0	611	0.531	4%
`price_10`	px	−269.5	2002	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	–	0.0	0	–	–
`ExpNet_lgbm`	px	−73.8	537	–	6%
`ExpNet_xgb`	px	−57.2	70	–	9%
`lgbm_weighted`	pb	−65.9	2	0.497	0%
`xgb_weighted`	px	−90.5	30	0.631	0%
`rf_balanced`	px	−89.2	11	0.467	0%

px = `price_only`; pb = `price_plus_book`; all = `price_book_settle`.

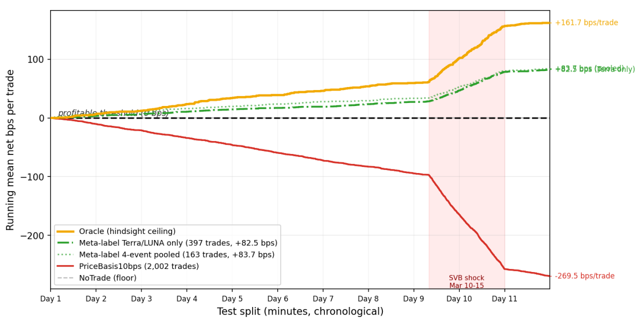

Figure 5: No calm-trained model ever crosses zero. Meta-labeling trained on Terra/LUNA is the only approach that achieves positive returns, earning +82.5 bps against the oracle's +161.7 bps. All other learned models lose money despite sometimes high AUROC. Oracle (gold); meta-label Terra/LUNA (green solid); meta-label 4-event pooled (green dotted); PriceBasis10bps (red); NoTrade (grey dashed). Shaded region: SVB shock window.

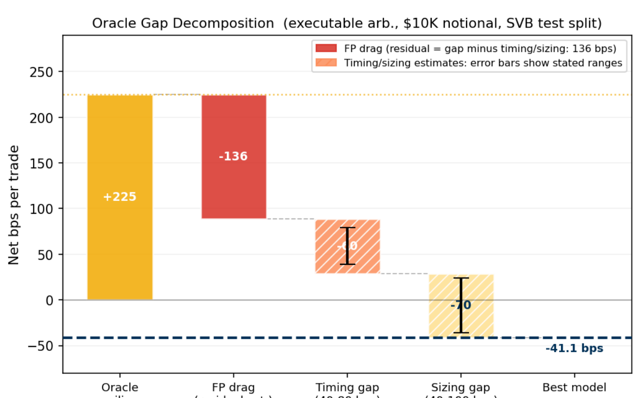

Figure 6: Oracle gap decomposition ($10K notional, SVB test split). False-positive drag is directly measured from false-positive trades. Timing error (40–80 bps) and sizing error (40–100 bps) are estimated from entry-delay and notional-scaling analysis.

drag: route-direction mismatch accounts for 581 of 1,851 false positives from price-rule trading. The remaining gap splits into timing error (40–80 bps, from entering 1–2 minutes late in a declining depth regime) and sizing error (40–100 bps, as depth depletion shrinks the executable fraction from 2.88% at $10K to 1.62% at $50K). Meta-labeling's 51% hit rate versus 6% for the price rule substantially reduces route-direction mismatch, though a 49% false-positive rate and roughly 113 missed profitable windows confine net returns to +82.5 bps. The prediction gap is not a model-selection problem. It is a training-distribution problem. No calm-trained model can bridge it. The only thing that works is letting the model see a different crisis first, which is where cross-mechanism transfer enters.

Table 9: Failure mode diagnosis by model family. All families fail for the same root cause: distributional mismatch.

Family	Failure mode	Key evidence
Price rules	Route mismatch	581/1,851 FP: USDT route dislocated, USDC at −38.7 bps
Tabular ML	Degenerate threshold	LightGBM: 0 trades; no threshold yields ≥25 with positive return
Sequence (GRU)	AUROC–P&L inversion	AUROC 0.80 (best sequence), net −239 bps (worst); 401 bps gap
RL (PPO-GRU)	Timing mismatch	919 trades at −29.2 bps; Terra pattern fires on wrong SVB windows
Regime detection	Mean-reversion violation	BOCPD AUROC 0.229; CEX basis reverts, not shifts

6.3 Cross-Mechanism Transfer

Cross-mechanism transfer means training on one type of crisis and testing on a completely different one. Here, the filter trained on Terra/LUNA's algorithmic collapse is applied unchanged to SVB's reserve-bank shock. This approach is the only method that crosses zero, earning +82.5 bps (51.0% oracle capture on the `basis_usdc_1m` task, oracle +161.7 bps from Table 5). A reinforcement learning agent given the same 17% primary-signal density earns −29.2 bps. The 111.7 bps gap points to explicit binary labels as the ingredient that reward optimisation cannot replicate at sparse positive rates. We treat this as a hypothesis for the multi-mechanism RL experiment, not a causal claim (Section 6.3.2).

6.3.1 Cross-Mechanism Transfer and Training Diversity. Meta-labeling achieves positive returns on every non-degenerate training condition tested (Table 10). Four-event pooled training spans algorithmic, exchange-credit, and regulatory mechanisms and reaches +83.7 bps (51.8% oracle capture).

Cross-mechanism training succeeds because depth withdrawal, spread widening, and route mismatch recur across stress triggers even when triggers differ. Figure 7 shows this directly: ask-side depth collapses and spread widens identically in primary-signal windows across Terra/LUNA and SVB despite mechanistically distinct causes. The order book responds to uncertainty, not its source. SHAP analysis (Table 11 and Figure ??) confirms the two-stage structure: price-basis features supply the candidate signal (94.8% of importance), while ask-side depth and spread filter execution quality (3.4%) once the model has seen stress-regime depth withdrawal.

Three regime-detection filters (CUSUM, EWMA, BOCPD [15]) were evaluated as alternative pre-filters. BOCPD achieves AUROC 0.229 on CEX cross-quote data. CEX basis is mean-reverting rather than change-point structured, so changepoint methods cannot substitute for the training-distribution fix that meta-labeling provides.

Table 10: Transfer and policy results (`basis_usdc_1m_gt10bps`, `price_plus_book`, SVB test split). Mean net bps per triggered trade; single-event block calibration throughout. DGP robustness ranges (500-seed bootstrap): Terra [78.7, 86.2], Celsius [74.1, 89.9], Pooled [75.3, 93.1].

Training source	Mechanism	Net bps	Trades	Oracle cap.
Oracle	hindsight	+161.7	316	100%
Calm-control	none	0.0	0	0%
Terra/LUNA only	algorithmic	+82.5	397	51.0%
Celsius/3AC only	exch. credit	+79.7	221	49.1%
FTX only*	exch. credit	+36.5	1	22.5%
Four-event pooled	alg.+exch.+reg.	+83.7	163	51.8%
PPO-GRU conditioned	RL on Terra	−29.2	919	−18%‡

*FTX: signal intensity < 5 bps; one trade is not interpretable as a transfer result.
‡Negative capture means the agent loses money relative to the NoTrade (0 bps) floor.

Table 11: Top SHAP importances, meta-labeling secondary classifier (LightGBM, Terra to SVB). Basis dominates; book features filter execution quality within primary-signal windows.

Feature	Importance	Group
USDC cross-quote basis	49.6%	price
Primary basis	32.8%	price
Deviation from $1	10.1%	price
USDT cross-quote basis	2.0%	price
Ask depth (10 bp)	0.5%	book/frag
Bid-ask spread	0.5%	book/frag
Price/basis total	94.8%	
Book/fragmentation total	3.4%	
Settlement total	1.8%	

The 50% saturation. All four training conditions produce oracle capture in the range 51.0–51.8% (Figure 8). Pooling four mechanistically distinct events does not lift this range. In the minutes immediately before a dislocation closes, ask-side depth partially recovers and the basis compresses, making late-stage windows microstructurally indistinguishable from genuine opportunities. The secondary classifier cannot resolve this ambiguity regardless of training diversity. We report this as an observed saturation, not a proven ceiling. Bootstrapping the four capture rates produces overlapping 95% intervals, which places a plausible true value near 50% but does not rule out higher capture with substantially more training diversity. Closing the gap further requires early-entry timing labels or a reactive simulator that models the agent's own market impact.

6.3.2 RL Policy Failure. This is a controlled comparison with one varying factor: how the profitable-window signal is delivered. Meta-labeling receives explicit binary labels (y=1 for each executable window); the conditioned PPO-GRU receives a sparse reward signal over the same windows. Both see identical training data at the same 17% primary-signal density. Despite this, the two methods differ by 111.7 bps in net return. The most parsimonious explanation is that explicit binary labels carry information about which windows survive execution that a sparse reward signal cannot encode. We treat this as a hypothesis. A definitive test requires multiple RL runs across the pooled multi-mechanism protocol, which we leave as the most immediate next experiment.

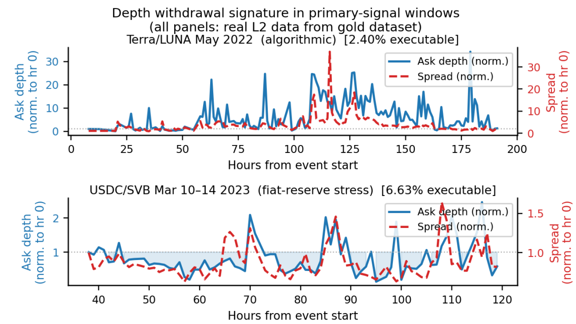

Figure 7: Ask-side depth (blue) collapses and spread (red) widens identically in primary-signal windows for Terra/LUNA (algorithmic collapse, May 2022) and USDC/SVB (reserve-bank shock, March 2023). The order book responds to uncertainty, not its cause. This shared microstructural signature is why cross-mechanism transfer succeeds.

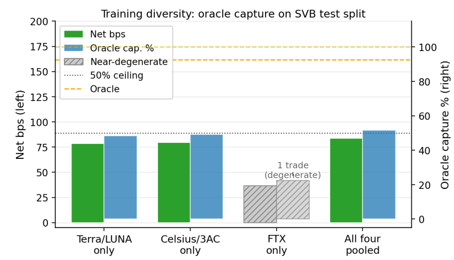

Figure 8: Oracle capture (%) across four training conditions. All four saturate in the 49–52% range (dashed line). FTX (hatched) is near-degenerate (one trade; shown for completeness but not interpretable as a transfer result).

Without conditioning, the PPO-GRU degenerates to zero entries: the 2.3% positive rate in the full Terra/LUNA stream is too sparse to calibrate the reward signal. Conditioning on primary-signal windows resolves the density problem, but the agent still earns −29.2 bps on 919 trades. The overtrading pattern points to a timing-transfer mismatch: the agent learned Terra/LUNA's depth-withdrawal sequence and fires on it in SVB even when the timing does not match.

7 Limitations

Tier-A coverage. VWAP execution labels (Tier A, meaning route-complete with a hindsight oracle) currently exist only for USDC/SVB and its recovery window. Extending to exchange-credit or DeFi events requires route-complete L2 archives not yet available. This is the primary bottleneck for expanding the benchmark's execution-grade scope.

Kline-proxy buy leg. The BTC-USDC buy leg uses kline-proxy depth for the SVB window. A 20% depth haircut widens the optical-to-executable ratio to 14× and keeps the oracle above +130 bps, confirming the headline result is not sensitive to this approximation.

RL market-impact feedback. The conditioned PPO-GRU's −29.2 bps may reflect a timing-transfer mismatch rather than a fundamental RL ceiling. Training on historical replay without market-impact feedback also miscalibrates sizing in thin-book regimes: the agent sees depth as it was, not as it would be after its own orders clear liquidity. Integrating the cross-mechanism initialisation protocol into a reactive simulator addresses both issues and is the natural next step toward closing the 111.7 bps gap between meta-labeling and the RL agent.

Release constraints and data ethics. The benchmark releases derived features, labels, event windows, source-tracking metadata, scripts, and frozen paper outputs under CC-BY 4.0. Some raw exchange order-book archives are API-limited, delayed, or subscription-gated. StressBench records route provenance and depth-source tags so future users can distinguish reproducible derived claims from raw-data acquisition constraints. No private user-level trading data is included.

8 Conclusion

Stablecoin arbitrage benchmarks built on price labels measure the wrong object. Across the SVB crisis, 34.3% of minutes appeared dislocated but only 2.88% survived VWAP book-walking, fees, and settlement—a 12× gap, replicated independently on Terra/LUNA at 5.9×. The two gaps have separate cures: Gap 1 is closed by execution-grade labels, and Gap 2 by cross-mechanism supervision, which succeeds because the order book responds to uncertainty rather than its cause. A single transfer from an algorithmic collapse to a bank-reserve shock recovers half the return available to a hindsight oracle, and pooling four mechanism classes confirms the result.

StressBench gives practitioners a corrected denominator: a backtest reporting AUROC or F1 on cross-quote basis labels overstates tradeable performance by roughly 12×, and those scores should be discounted before any capital decision.

We see three directions forward. First, *coverage*: extending Tier-A execution labels to exchange-credit and DeFi events turns cross-mechanism transfer from a two-event result into a general law. Second, *reactivity*: a market-impact simulator initialised with the cross-mechanism protocol is the direct test of whether the 111.7 bps gap between meta-labeling and reinforcement learning closes once the agent prices its own footprint. Third, *timing*: early-entry labels target the late-stage ambiguity that currently caps oracle capture near 50%. The oracle ceiling supplies a principled performance target for all three. We release the catalogue, labels, oracle, and harness under CC-BY 4.0 so that any model claiming to capture stablecoin stress arbitrage can be evaluated against the correct denominator.

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